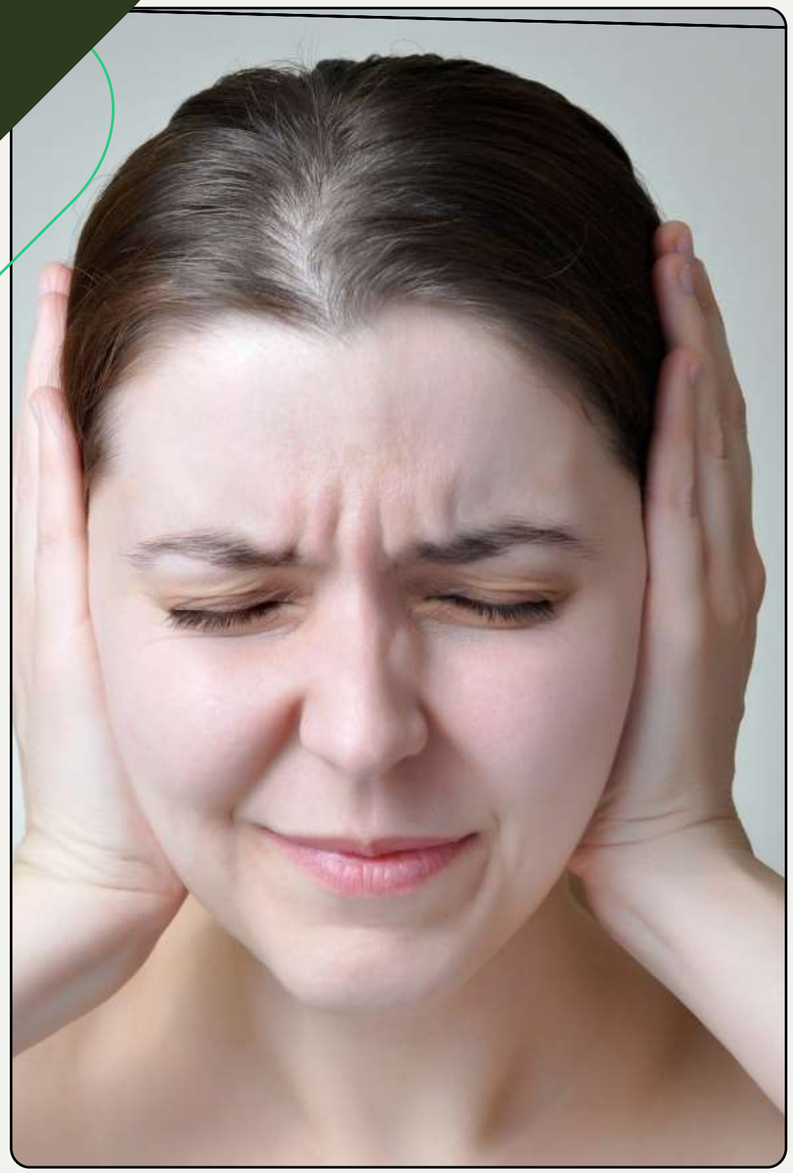


Soundscape-BD: A Dataset of Urban Sound Pollution Collected Across Dhaka

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Team SuperNovas



Overview of Sound Pollution in Urban Bangladesh

Introduction to Sound Pollution

Exploring the effects of noise in urban areas

Major Sources of Noise Pollution

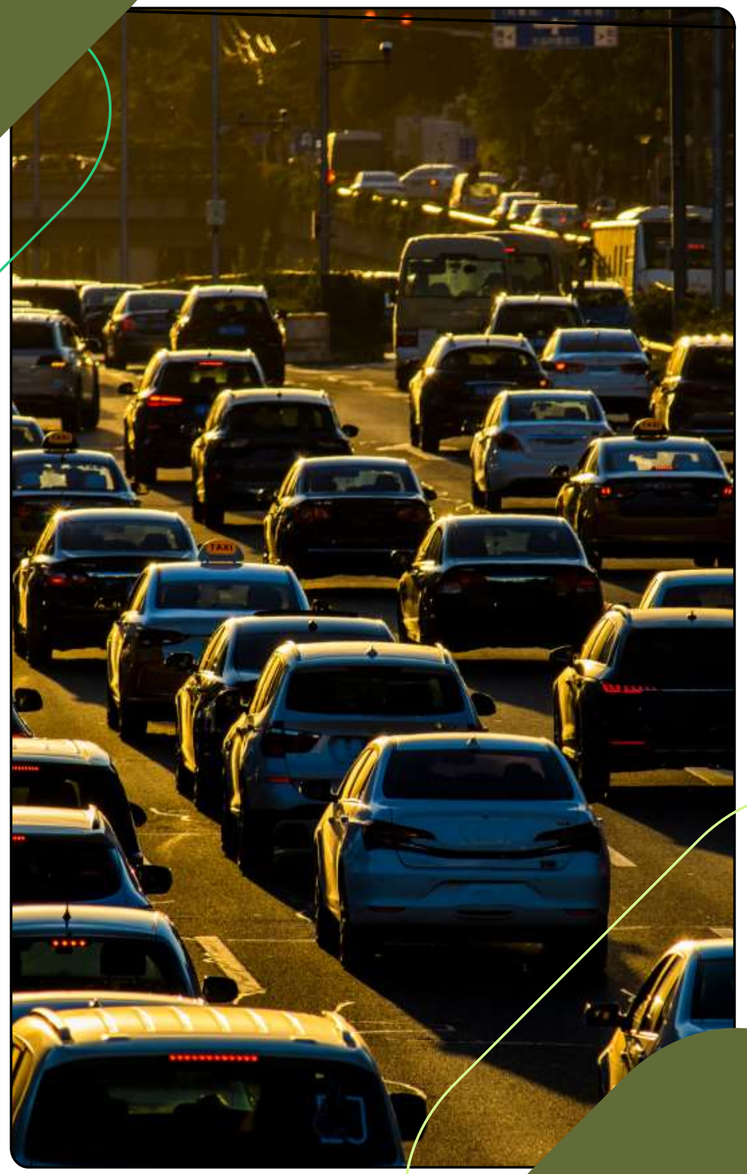
Identifying key contributors such as traffic and industry

Health Impacts of Noise Pollution

Discussing physical and mental health consequences

Detection and Monitoring Methods

Reviewing techniques for measuring noise levels



Major Sources of Sound Pollution in Bangladesh

Traffic: The Most Prominent Source

High vehicle density contributes significantly to noise levels.

Construction: Rising Challenges in Urban Areas

Ongoing projects add to the cumulative sound pollution.

Industrial Noise: A Growing Concern

Factories and industries play a major role in sound pollution.

Events: Temporary but Loud Disturbances

Public gatherings and celebrations generate elevated noise levels.

Sound Polution Detection Across Urban Areas

High Traffic Areas

Traffic noise often exceeds **WHO recommended limits** in major cities.

Construction Zones

Construction activities contribute significantly to **urban sound pollution levels**.

Industrial Areas

Industries frequently operate above **WHO noise thresholds**, causing health concerns.

Public Events

Large events can spike noise levels, often **exceeding regulatory limits**.

“Sound pollution can lead to serious health issues, including stress, sleep disturbances, and cardiovascular problems. Addressing these effects is crucial for improving urban living conditions.”

– WORLD HEALTH ORGANIZATION

Literature Review



Paper Title	Key Methods / Models Used	Relevance to Our Project	Identified Gap
Urban Sound Classification Using CNNs	Convolutional Neural Networks (CNN) on Mel-spectrograms	**Effective for urban sound classification including vehicles and construction sounds**	**Does not specifically address mixed or overlapping sounds common in dense traffic environments like Bangladesh**
Deep Learning for Acoustic Scene Classification	**CNN, RNN, and attention mechanisms on spectrograms**	**Useful for detecting complex soundscapes including mixed classes and background noise**	**Limited focus on emergency vehicle sounds (ambulance, police) and rare classes**
Vehicle Sound Classification Using MFCC and SVM	**MFCC features + Support Vector Machine (SVM)**	**Traditional approach for vehicle sound classification (car, bike, bus)**	**Struggles with noisy environments and difficult classes like trucks and emergency vehicles**
Sound Event Detection in Urban Environments with DNN	**Deep Neural Networks (DNN), data augmentation**	**Addresses multiple urban sound classes, including construction and factory sounds**	**Limited dataset diversity for South Asian urban sounds; emergency vehicle sounds underrepresented**



Step 1

Collect **audio data** using A100 microphone



Step 2

Audio Prepocessing



Step 3

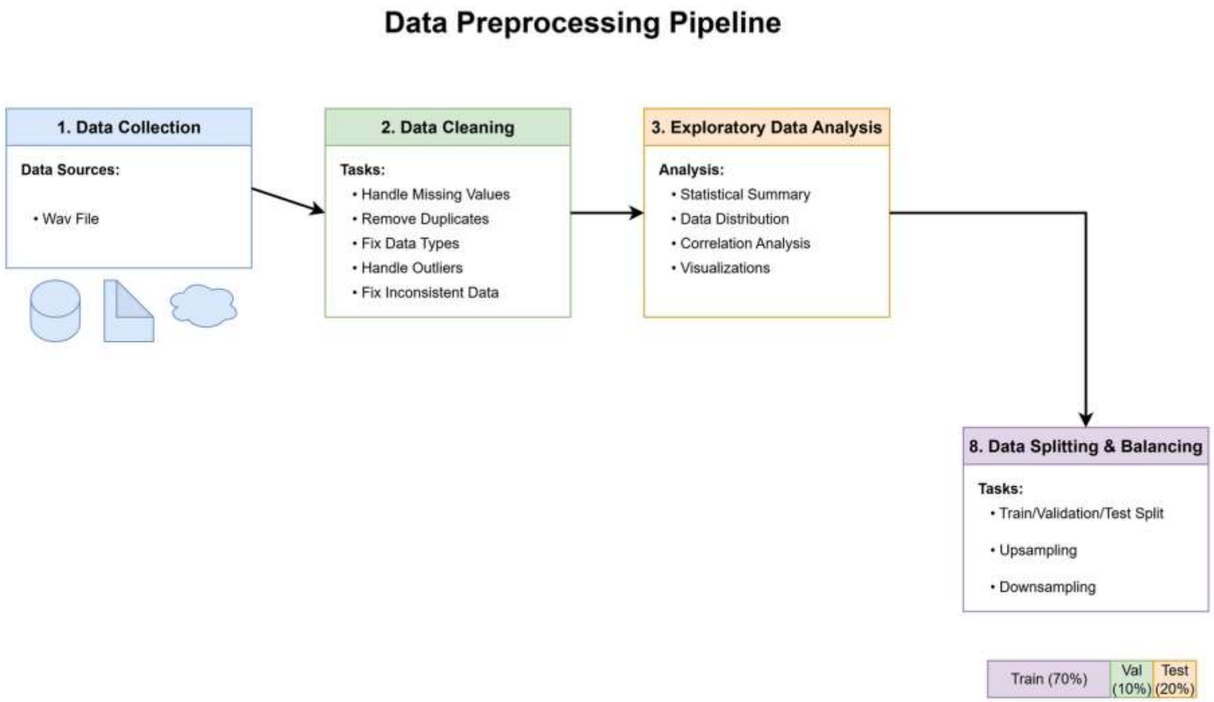
Train

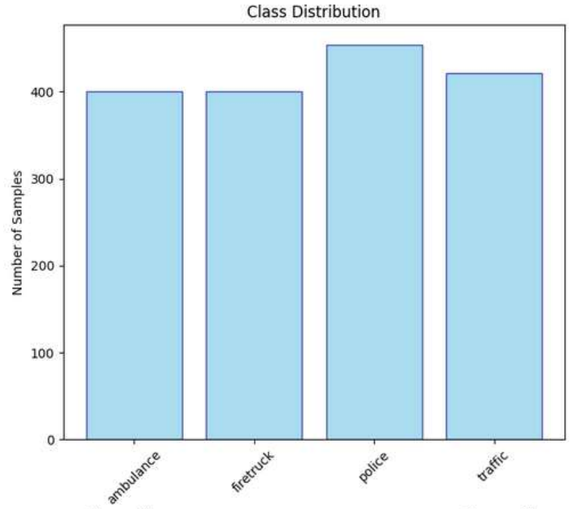
Data Acquisition

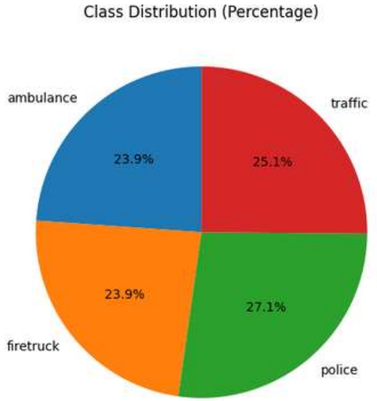


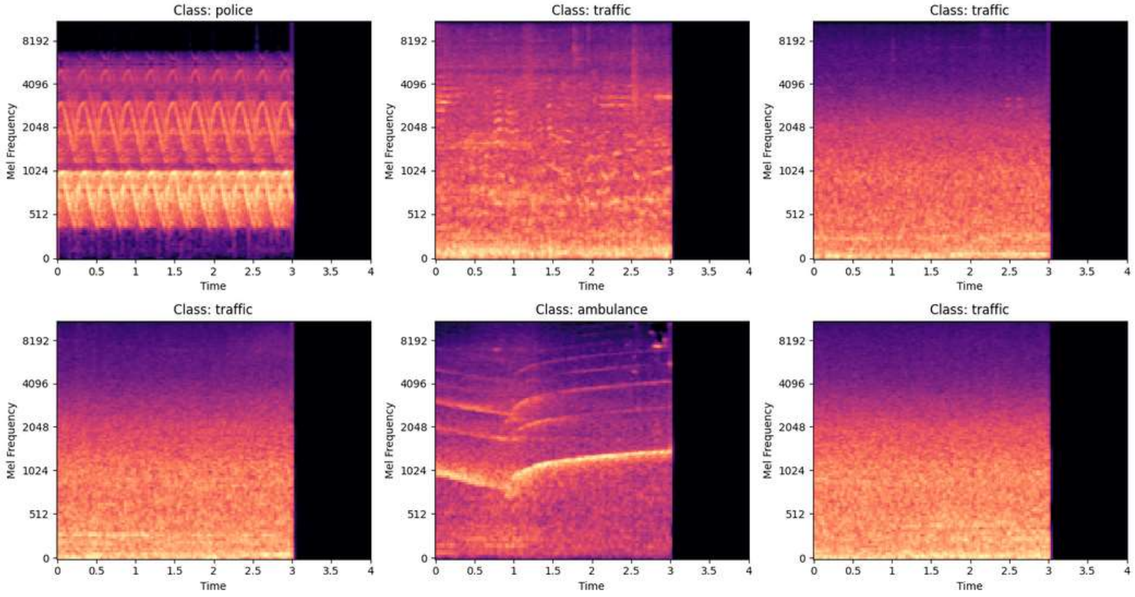
Component	Specification
Subjects	Vehicles + 2 m distance
Audio	A100 microphone, 16 kHz, mono, WAV
Location:	Several Places in the Data City

Prepocessing Steps

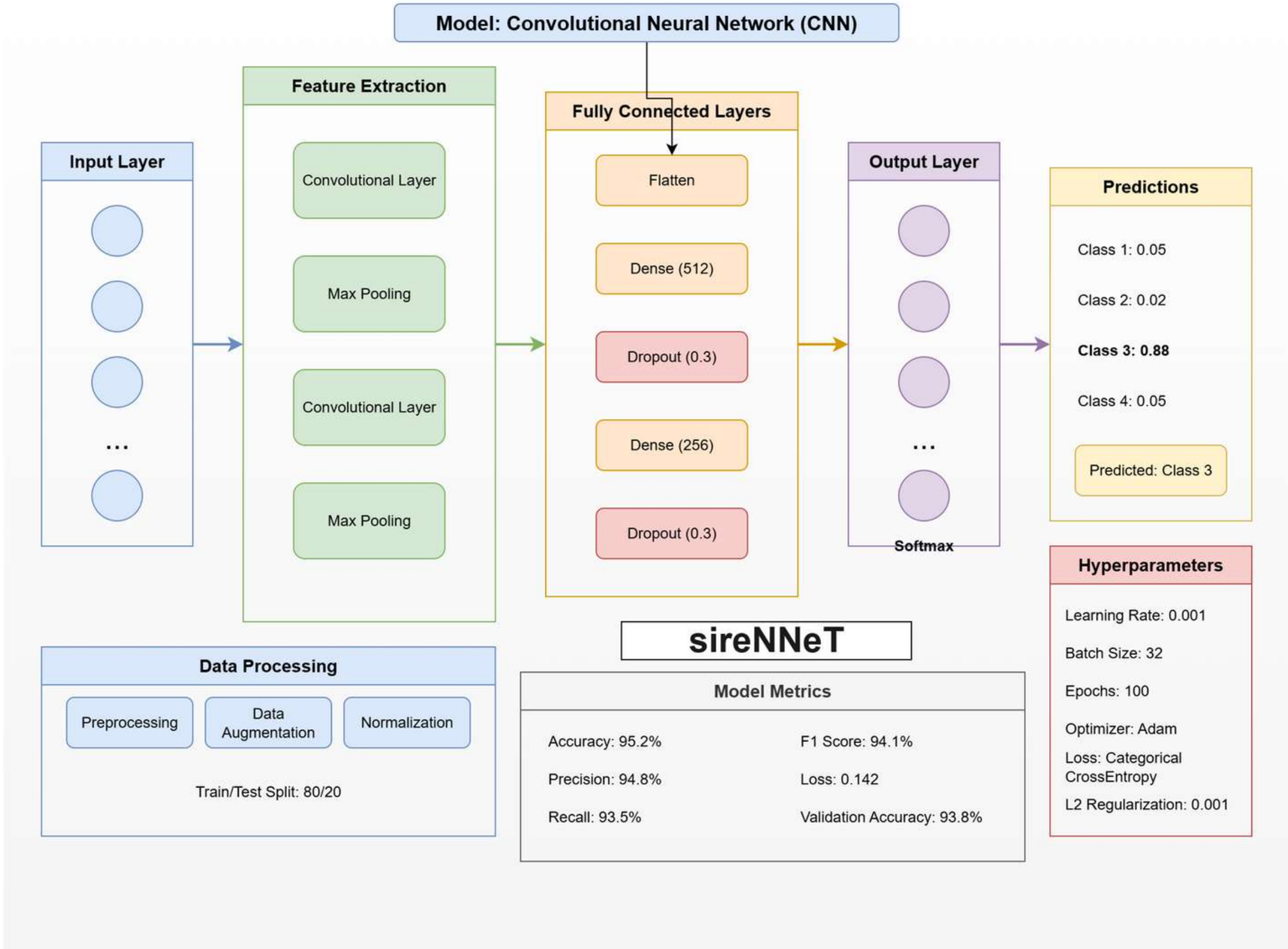




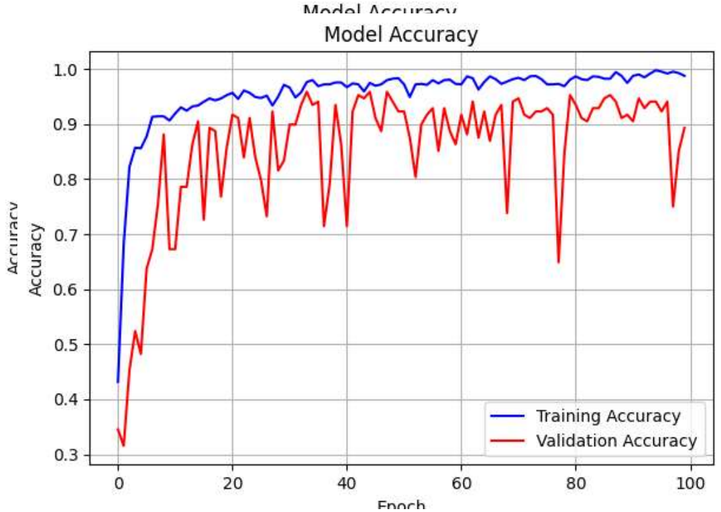
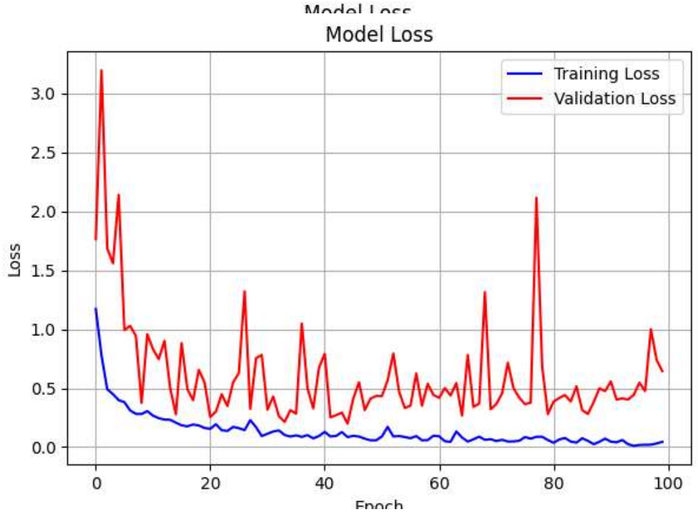




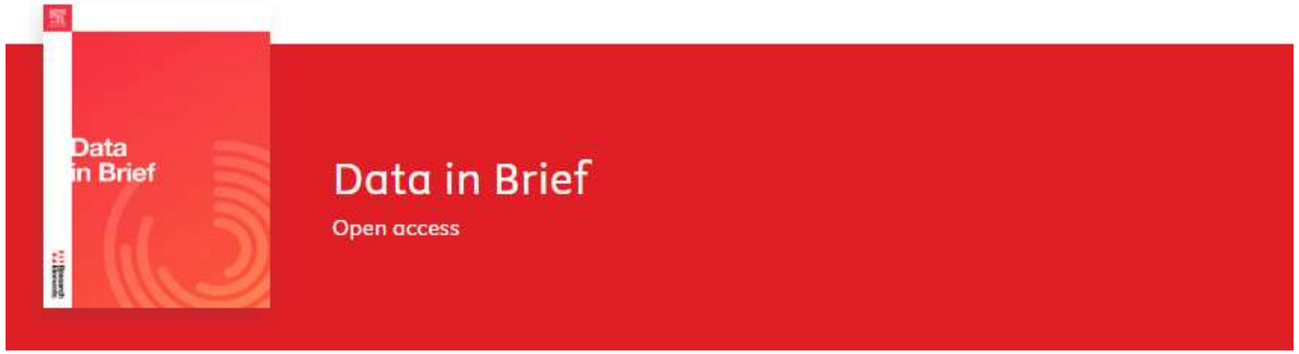
Methology


Model: Convolutional Neural Network (CNN)
Input Layer
Feature Extraction
Convolutional Layer
Max Pooling
Convolutional Layer
Max Pooling
Fully Connected Layers
Flatten
Dense (512)
Dropout (0.3)
Dense (256)
Dropout (0.3)
Output Layer
Softmax
Predictions
Class 1: 0.05
Class 2: 0.02
Class 3: 0.88
Class 4: 0.05
Predicted: Class 3
Hyperparameters
Learning Rate: 0.001
Batch Size: 32
Epochs: 100
Optimizer: Adam
Loss: Categorical CrossEntropy
L2 Regularization: 0.001
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Data Processing
Preprocessing
Data Augmentation
Normalization
Train/Test Split: 80/20
Model Metrics
Accuracy: 95.2%
F1 Score: 94.1%
Precision: 94.8%
Loss: 0.142
Recall: 93.5%
Validation Accuracy: 93.8%

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Class Name	Evalution Meteries
1. Ambulance: 400 files 2. Police: 454 files 3. Firetruck: 400 files 4. Traffic: 421 files	 

Our Custom Datasets Aim



Class Name
1.Rickshaw-1000 2.car-1000 3.Bike-1000 3.Construction 4.CNG-1000 5.Bus -1000 6.track(difficult)-1000 7. Ambulance(difficult) -1000 8.Police(difficult)-1000 9.Factory sound(difficult).-1000 10.mixed class-1000



Thank You!